DevOps Mock Interview Question Bank

GIT:

Git Fork vs Git Clone

Explain a scenario where you used Git Fork instead of Git Clone.

Git Fork in action with example

Create a fork of the project mentioned below and create a pull request

Git Fetch vs Git Pull

Show how does Git Fetch and Pull work in Realtime.

Which command do you use mostly - Git Fetch or Git Pull and why ?

Practice Git Fetch vs Git Pull on a GitHub Repo.

Git Rebase vs Git Merge Detailed Explanation

Show practically how Git Rebase is different from Git Merge

How to explain Git Merge vs Git Rebase in Interviews (Short Explanation)

Practice Git Merge vs Git Rebase

Explain the Git Branching Strategy that you used in your company.

Explain 3 challenges that you faced with Git during your work experience.

Explain the recent challenge that you faced with Git and how did you address it

How do you handle Merge conflicts in Git ?

Explain Git Merge Strategies - Ours and Theirs strategies

Create a Merge Conflict and address it locally using Git merge

Have you ever used Git tags ? If yes, why ?

How do you combine Multiple commits into a Single commit ?

Explain 10 Git Commands that you use on Day to Day basis ?

I want to ignore pushing changes to a file to Git, How can you do it ?

What is the purpose of .git folder in a Git repository ?

Can you restore a deleted .git folder ?

A teammate accidentally committed a Kubernetes Secret (base64 encoded) to Git.

LINUX:

10 linux commands that you use on day to day basis.

Can you restore lost pem file ? If not how can you access the instance ?

/Var is almost 90 percent full, What will be your next steps ?

Linux Server is slow due to High CPU utilization. How will you fix it ?

Application deployed on Nginx returns Connection Refused, How will you fix it ?

SSH to an instance stopped working ? How will you troubleshoot the issue ?

Find and list the log files older than 7 days in /var/log folder.

Find and remove the log files older than 30 days in a folder.

Cronjob + Shell script to perform advanced log rotation (Scenario Provided)

Bulk creation of linux users using CSV file.

Service Health monitor script in Bash

Find and delete files over 100MB

Get the list of users who logged in today (scenario - some packages deleted)

Website doesn't load, How will you investigate ??

Using sed command, how to remove first and last line of the file

What are the different types of variables in Linux ?

Kill vs Kill -9 in Linux

Networking:

Explain DNS in simple words.

Explain the complete flow of the request from client to server (OSI Model)

Explain the difference between Forward Proxy and Reverse Proxy

User Reports slowness in the app, How would you approach this ?

Curl Works with IP, Fails with Domain, Why ?

Website returns 502 HTTP Status code. What can be the issue ?

[Many people cant answer] What is the difference between 0.0.0.0 and 127.0.0.1 ?

What is the difference between Public and Private Subnets ?

You accidentally created a private subnet instead of public.How will you fix it?

CI/CD:

What are Jenkins shared libraries and how do they work?

Talk about 5 build targets that you use on day to day basis.

Which artifact repository do you use for builds ?

How do you configure Artifactory for your application in Maven ?

Build Passed Locally but Fails in CI, how will you troubleshoot ?

CI Pipeline Succeeds but App is Broken in Prod, What action will you take ?

Pipeline Slows Down Over Time (Builds taking more time) - How will you fix ?

A developer pushes a feature branch, but the pipeline doesn’t trigger.

Your build fails because it can’t download a dependency from your artifact repos

Python Build Fails on CI But Works Locally, What can be the issue ?

Explain Python Application Build Process in Detail.

Using Static Code Analysis what kind of problems can you identify ?

Static Code Analysis Slows Down CI Pipeline, How will you fix it ?

App in ‘OutOfSync’ State in Argo CD, But No Git Changes

When a build fails in Jenkins, How will you send an email ?

Terraform:

What is the difference between for_each and for in Terraform?

What are modules in Terraform and why should we use them ?

What is the role of statefile in Terraform ?

Have you considered storing statefile in Git instead of AWS S3 or Azure Blob ?

Explain Terraform Statefile Management

Two DevOps Engineers attempts to update statefile at once. What happens ?

We don't have a cloud account. Where can we store the statefile?

Do you use Terraform Enterprise or Community version ?

Have you heard about opentofu ? Do you think it is better than Terraform ?

Write terraform code to create any resource on AWS ?

What is the difference between Resource and Datasource in terraform ?

Docker:

Docker Container Exits Immediately, how will you troubleshoot ?

[90 Percent get this wrong] What is the purpose of EXPOSE in Dockerfile ?

Port is Not Accessible on localhost even after Port mapping in Docker

Data lost when container stops and restarts, How will you fix it ?

You made change in your code, rebuilt the image, but the change isn't reflected?

App Crashes with "Permission Denied" in Container but works fine on localhost.

Docker host is running out of disk space. How do you clean up?

How will you Debug a Live Container ?

Which container registry do you use in your organization ?

Explain the difference between CMD and Entrypoint in Docker.

What docker commands you use on day to day basis ?

when will you forcefully remove a container and how ?

Kubernetes:

Explain Kubernetes Cluster Architecture.

How various components of Kubernetes interact when you run kubectl apply (Pod)

What is the purpose of services in Kubernetes ?

Why is hardcoding Pod IP communication is a bad practice ?

What are the types of services in Kubernetes ?

What are labels and selectors in Kubernetes ?

What would you recommend NodePort service or Load balancer type service and why

How Kubernetes Services are related to Kube Proxy ?

What is the disadvantage of LoadBalancer service type ?

What is a Headless Service in Kubernetes and when did you use it ?

Can a Pod access Service in different namespace, if Yes, how ?

Explain how you can restrict access to db pod to only one app in the namespace ?

Explain the deployment strategy that you follow in your organization.

Explain the rollback strategy that you follow in your organization.

Design a solution to avoid rollbacks.

Explain the deployment strategies that you used in the past.

Explain the role of CoreDNS in Kubernetes.

A DevOps engineer tainted a node as "Noschedule". Can you still schedule a pod ?

Pod is Stuck in CrashLoopBackOff, What steps will you take ?

what is the difference between liveness and readiness probes ?

Explain the difference between Ingress and LB service type.

Your app works with ClusterIP but fails with Ingress. How do you troubleshoot it

Why do I need to setup Ingress controller after creating ingress ?

We have an inhouse load balancer, Can we use Ingress with our load balancer ?

Your Deployment has replicas: 3, but only 1 pod is running. What could be wrong?

Your pod mounts a ConfigMap, but changes to the ConfigMap are not reflected ?

Explain how Node Affinity works and when will you use it ?

What is the difference between Node Affinity and Node Label Selector ?

What is container runtime in Kubernetes ?

What is Kubernetes QOS ?

What are requests and limits in Kubernetes ?

Explain 3 challenges that you faced while working on Kubernetes ?

Can we use Kubernetes Master for scheduling the pods ?

Explain Horizontal vs Vertical Scaling in Kubernetes or in general

What are the different types of secrets in Kubernetes ?

observability:

What is the difference between monitoring and observability ?

how to emit custom logs and metrics in your application ?

What kind of metrics do you scrape with prometheus in your current organization?

Have you worked on Observability if yes, explain what did you do ?

What is the difference between logs, metrics and traces ?

What is the difference between push and pull based monitoring ?

Which tools have you used to build observability stack ?

Users report slowness in app. Logs don’t show errors, and CPU is good. Fix it.

How do you trace a request across multiple microservices in a Kubernetes cluster

A pod crashes randomly with OOMKilled. How do you identify and fix this?

You got woken up at 2AM by false alarms. What's your strategy to reduce noise ?

AWS:

Explain how will you design a highly available and scalable multi-tier app.

What is AWS NAT and when is it used ?

How to enable Internet access to the application deployed in a private subnet ?

Can applications in different subnets of a VPC interact by default, If no, why ?

Explain NACL vs SG and which on do you use in your ogranization ?

EC2 Instance terminated unexpectedly, How will you troubleshoot ?

Lambda Function Fails Randomly, How will you fix the issue ?

What will you do when AWS RDS Storage is Full ?

Developer Deleted Critical Resources like S3, RDS and EC2. What will you do ?

Explain a cost optimization activity that you performed in the current org.

Explain a recent challenge that you faced with AWS and how did you solve it ?

Auto Scaling Group Not Launching EC2, What can be the issue ?

Which AWS services do you use in your day to day life ?

Have you used AWS EFS ? If yes, what issues did you run into ?

When will you go for EFS over EBS in realtime ?

Disable AWS console access to the IAM Users.

How can AWS lambda in one AWS account connect to s3 bucket in other account ?

What is AWS STS and how does it work ?

What is trust policy in AWS and why is it used ?

How a Lambda function in AWS Account A interact with Dynamodb in Account B ?

What are the disadvantages of using EBS volumes in Kubernetes ?

AWS Secret Manager vs Parameter Store

Explain your day-to-day activities related to Databases

Have you worked with Lambda in AWS ? Explain your activities with lambda.

What is the difference between IAM user & role ?